

Daimler Truck North America

Work Term Assignment for E S 125N

2nd Term

Noah Hickman

Department of Mechanical Engineering

The University of Texas at Austin

Summer 2025

Table of Contents

List of Figures and Tables	3
List of Acronyms	3
Introduction	4
Work Environment	5
Technical Summary	
Voice Trigger Transcription and Indexing	6
Assisting with Product Testing	10
Final Conclusion	10
References	12

List of Figures and Tables

Figure 1: GoDAP App Window

Figure 2: Custom Vector Logging Setup

List of Acronyms

DTNA - Daimler Trucks North America

PVE - Product Validation Engineering

PTM - Product Testing: Mechatronics

GoDAP - Go Daimler Analysis Platform

ADAS - Advanced Driver Assistance Systems

DTICI - Daimler Truck Innovation Center India

CAN - Controller Area Network

ECU – Electronic Control Unit

I. Introduction

Daimler Trucks North America (DTNA) is the leading commercial vehicle manufacturer in North America, originally founded in 1940 as Freightways Manufacturing Company. The company became Freightliner before its 1981 acquisition by Daimler-Benz AG, later becoming DTNA, a standalone subsidiary under Daimler Truck AG in 2021 after corporate restructuring (Daimler Truck North America, n.d.). With its Freightliner, Thomas Built Buses, and Western Star commercial vehicle brands, along with its Detroit Diesel engine brand, DTNA has garnered more than 43.5% of the Class 6-8 vehicle market in North America as of 2023 (Greenhalgh, 2023).

Each distinct brand helps Daimler offer a plethora of vehicles for specialized use cases, such as sleeper long-haul, regional day cabs, and even specific vocational vehicles like school buses and rugged haulers. DTNA is headquartered in Portland, Oregon, with many other manufacturing plants across the U.S., as well as Mexico. With over 28,000 employees across North America, DTNA's mission focuses on delivering customer-first innovations that are "efficient & electric; safe & automated; reliable & connected". Additionally, they seek to truly solve problems that develop technologies without compromise, with sustainable solutions being one of their primary goals (Daimler Truck North America, n.d.).

I worked at DTNA's Product Validation Engineering (PVE) campus in Portland as part of the Product Testing: Mechatronics (PTM) department. Specifically, I worked with the data analytics team, along with three other full-time engineers, who are responsible for developing the department's internal data analysis platform, GoDAP. My project focused on adding voice transcription and tagging features to GoDAP. I was one of 5 interns within my department, with over 70 fellow interns located in Portland over the summer.

II. Work Environment

At DTNA, I worked closely with my GoDAP team members, including my primary mentors, Terrell Olvera, A.J. Marchese, and Pat Barton. Each of them brings their own specialties to our team, with Terrell having extensive knowledge over both the hardware and software sides of vehicle data collection, A.J. being primarily focused on DevOps, and Pat concentrating on data analytics and cloud computing. Additionally, I had the opportunity to work with other teams across the PTM department, such as the Brakes and Advanced Driver Assistance Systems (ADAS) groups, during department over-the-road and feature testing events.

As part of the GoDAP team, I attended daily morning scrum meetings, which followed a round table format, detailing the current roadblocks, goals, and deliverables for each team member's project. Furthermore, my team had bi-weekly meetings with our remote software development team at Daimler Truck Innovation Center India (DTICI) for detailed software development discussions, as well as a weekly Systems team meeting for general updates, which consisted of the GoDAP team, the Brakes team, and our shared team manager, Dawn Rinehart. Throughout my co-op, I consistently asked my mentors for advice and guidance on my project and was able to present my progress to the engineers in the department by having them directly use my tools during over-the-road testing. Additionally, I got to present my project to the managers of my department at the midway point of the program, as well as during my final week, to the board members of the entire PVE division.

In addition, as a benefit to being a summer intern, I also had the opportunity to participate in intern tours and social events across the DTNA campus. These included a tour of the Western Star manufacturing plant, the autonomous truck divisions, and all PVE facilities, which featured extreme testing chambers and other cutting-edge tooling.

III. Technical Summary

Voice Trigger Transcription and Indexing

My primary project as a GoDAP team intern involved developing voice trigger transcription and indexing capabilities for the GoDAP platform, which would enhance the overall traceability and indexing of our data. This project aimed to extract meaningful data from each recorded trigger, reduce the need for taking external notes about each trigger during testing, and lay the foundation for more scalable search functionalities within the future of the app.

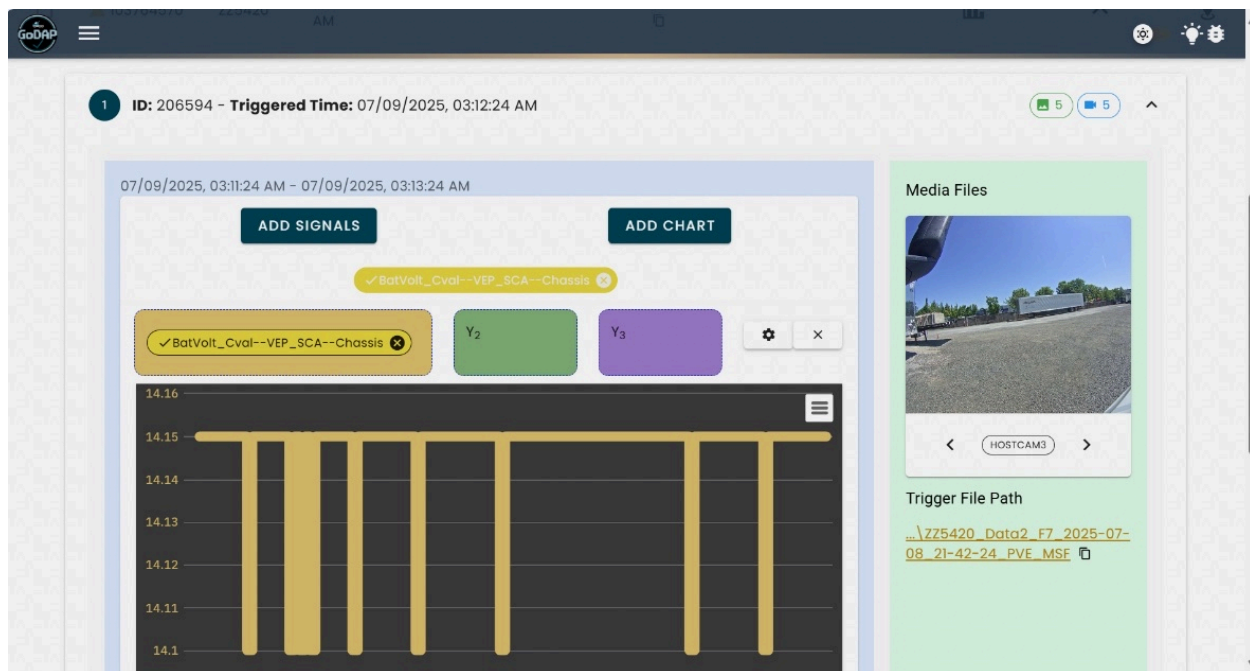


Figure 1: GoDAP App Window

For starters, the PTM GoDAP app is best described as a departmental internal data storage and analysis tool. GoDAP allows us to hold all previous and current testing data files, also known as data buckets, across hundreds of tested trucks, in which the data is recorded from the many controller area network (CAN) buses located on the trucks. This data holds key information, not only on common vehicle parameters across the engine and main vehicle electronic control units (ECUs), but thousands of electronic signals covering all the truck's

advanced electronic and mechatronic features, from safety to driving aids, and much more. Before each test drive, test engineers insert storage discs into Vector GL loggers onboard the trucks, which gives us continuous data acquisition throughout the entire drive cycle.



Figure 2: Custom Vector Logging Setup

Furthermore, an event trigger is a specific position in the data buckets where a particular event or situation needs extra attention during data analysis. This could be due to a system failure, a successful test, or any interesting truck performance that needs to be reviewed carefully in the future. To do this, engineers press the trigger button installed on the dashboard of the truck to record pictures around the truck and of the cab's dashboard, as well as a short voice recording from a microphone located in the cab, which creates the voice triggers my project was focused on. Despite the high volume of useful data collected, locating specific trigger events within multi-hour recordings is time-consuming and inefficient for test engineers, which potentially leads to spending more time trying to review data than being able to troubleshoot truck features.

My project and the GoDAP app as a whole help streamline that process and make finding important test data less of a hassle.

To develop this project, I coded custom voice transcription and sound detection scripts using Python, complete with advanced parsing and formatting algorithms for cleaner transcription results. I designed and implemented a PostgreSQL-based backend to store the transcriptions, keywords, and media tied to each voice trigger. With this backend architecture, I created an easy retrieval process for the front-end software engineers to integrate my project into the app's workflow. Utilizing both Python and PostgreSQL took up most of my day-to-day, as I had to continue building my foundational Python skills and begin self-teaching SQL as needed for this project. To perform the transcription of the voice triggers and extract useful data for indexing on the app's front end, I decided to employ a three-script package.

The first script involved analyzing each of the trigger packages, storing pictures, videos, and associated metadata in our database backend. Each audio file was transcribed using OpenAI's Whisper model along with the WhisperX library for alignment and word-level timestamping, leading to more accurate transcriptions from distorted or complex audio. I chose to utilize these libraries due to their powerful semantic accuracy and ability to maintain their accuracy if an audio file is particularly distorted or tricky. This challenging circumstance was often caused by an engineer using the radio, having a meeting in the background, or even having multiple engineers in conversation as the microphone records. After being transcribed and stored in the database, a second script would run in succession, parsing the transcription, looking to detect any keywords to tag the transcription and trigger with. The keywords used in this algorithm would be user-driven, where users of the GoDAP app can add any custom tag they would like found and marked in any transcription. I decided to implement this to allow

engineers to search specific test cases, features, or even names that occurred in the audio file of a trigger, which allows for narrowing down a trigger quickly, even without the use of the truck's name or recorded date. In addition, I utilized the Short-Term Fourier Transform (STFT) capabilities of the Librosa library in this script, creating a function that detects changes in audio frequency over time, giving my script the ability to detect certain Optical Acoustic Warnings (OAW) from the truck's cluster during a drive, extracting more metadata from the triggers. Lastly, a final script is utilized to scrub each transcription, remove names, censor any non-engineering content, and generate a cleaned transcription to be posted with its respective trigger on the website. To automate transcription filtering, I developed a custom-trained classifier model using a Bidirectional Encoder Representations from Transformers (BERT) model via the Hugging Face Transformers library, trained on more than 300 previous transcriptions. A classifier model like this helped solve the issue of removing transcriptions that were not relevant to the situation in the trigger, while keeping relevant transcriptions that didn't contain obvious keywords, with exponentially less complexity and hardware requirements compared to a large language model.

To deploy the code, I transferred the scripts, written locally on Visual Studio Code with Windows 11, to an offline, remote Linux-based server, using Docker images and deployment through Portainer. This was by far one of the most challenging aspects of the project, having to manage library dependencies while ensuring consistent runtime and environment replication between my local setup and the remote server, as my code stack would need to run constantly to process data as it came in, keeping the app up to date. To do this, I learned to optimize my Python scripts, condensing them and prioritizing system reliability and performance as I iterated through my programs. Not to mention, I strengthened my skills in source control and

documentation by utilizing GitHub for version management and creating detailed project notes to create a basis for future development on my additions to the GoDAP app.

Assisting with Product Testing

In addition to my primary project, I was also occasionally involved in data collection and testing projects throughout my department. For example, I took part in a three-day over-the-road test spanning parts of Oregon and Washington. This involved a multi-truck convoy, with test engineers focusing on features across the trucks, testing in live, real-world environments. The test also served as the initial deployment environment for the voice trigger system I had developed, allowing engineers to begin recording audio triggers in real driving conditions. The field recordings from this three-day trip became the first data I utilized to build and refine my scripting project.

Furthermore, since my PTM Systems group is shared with the Brakes team, this allowed me to occasionally assist in braking tests, which helped me gain an understanding of upcoming truck features and allowed me the chance to drive as a follow or chase car as needed.

IV. Final Conclusion

The past two semesters of work experience provided me with a refreshing break from classes while still allowing me to gain plenty of hands-on skills and professional habits that will help me in my future career. It has greatly changed my academic goals, as I am now more focused on growth and learning through trial and error, instead of chasing perfect grades or prioritizing coursework unrelated to my interests. Additionally, after diving into multiple different areas within the automotive industry, I now understand the value of expertise and having extensive knowledge in a certain niche, which has inspired me to potentially explore graduate school. I am more motivated to complete my degree now that I understand the inspiring challenges and jobs

within the world of engineering. My self-confidence increased, not just through diving into the unknown with two new jobs, but also by living in two completely different environments alone, being on the East Coast in the winter, and on the West Coast in the summer. One of my major achievements from these opportunities has been becoming a more well-rounded engineer, with real-world, multidisciplinary skills. I've gone from zero job offers my freshman year, to one in the spring, two this summer, and now I have the confidence and momentum to build my future. In terms of personal goals, the extra free time of a conventional work week, on top of being paid for my time, has enabled me to live a healthier life and start exploring my hobbies, new and old, with some of the most notable being weightlifting and solo traveling across both the U.S. and internationally. Academically and professionally, the skills I have gained not only by working in team settings, but also technical skills individually, will carry forward as I enter the full-time job market post-graduation, as well as in my projects and student organizations as I finish my degree. For the remainder of my undergraduate degree, I plan to focus on courses exploring my interest in multidisciplinary fields, such as mechatronics or systems and controls-related CGE courses, while continuing my involvement on the Longhorn Racing combustion team to further explore my automotive passions. Additionally, I hope to secure an internship position, ideally at another automotive or aerospace company within the next year, deep within a test engineering division, continuing to get more hands-on exposure and exposure to new technologies. Finally, I plan to apply to the integrated BSME/MSE program, focusing on dynamic systems and controls, which I feel will give me a great foundation to dive into a career in test engineering.

References:

Greenhalgh, K. (2023, May 9). *Daimler Truck Q1 Sales jump*. Transport Topics.

<https://www.ttnews.com/articles/daimler-earnings-q1-2023>

Daimler Truck North America. (n.d.). *History*.

<https://northamerica.daimlertruck.com/company/history>

Daimler Truck North America. (n.d.-b). *Purpose*. <https://northamerica.daimlertruck.com/purpose>